

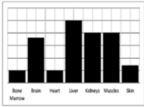
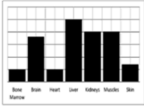
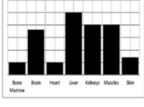
1.1.c. Energy for exercise

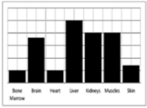
Learners will develop their knowledge and understanding of Adenosine Triphosphate (ATP) as energy currency, along with the principle of the coupled reactions and resynthesis of ATP. The detail of the different energy systems will be known and learners will understand the energy continuum and factors that affect the interplay of the energy systems.

Learners will be able to interpret figures relating to the contribution of the three energy systems to exercise of different intensities and durations.

The recovery process will be known, along with an understanding of interpretation of figures relating to the contribution of the three energy systems to exercise of different intensities and durations.

Learners will develop their knowledge and understanding of the effect of exercise intensity on excess post exercise oxygen consumption (EPOC) and implications of the recovery process for planning exercise or training sessions related to physical activities and sports.

Topic Area	Content
*Adenosine Triphosphate (ATP) and energy transfer 	- ATP as 'energy currency' - principle of energetically coupled reactions: - breakdown of ATP to ADP (Adenosine Diphosphate) + P (phosphate) - resynthesis of ATP from ADP + P.
*Energy systems and ATP resynthesis  	- energy systems: - ATP-PC (Phosphocreatine) system - glycolytic system - aerobic system - for each system: - type of reaction (aerobic or anaerobic) - chemical or food fuel used - specific site of the reaction - controlling enzyme - ATP yield - specific stages within the system - by-products.
*ATP resynthesis during exercise of differing intensities and durations  	- the energy continuum - predominant energy system used during exercise: - how intensity and duration of exercise influence which energy system is predominantly used to resynthesise ATP - interpretation of figures relating to the contribution of the three energy systems to exercise of different intensities and durations - interplay of energy systems during intermittent exercise and factors that affect this interplay - intensity of exercise - duration of exercise - recovery periods - fitness levels.

Topic Area	Content
*The recovery process  	• how the body returns to its pre-exercise state: ◦ Excess Post exercise Oxygen Consumption (EPOC) • fast components of EPOC, the processes that occur and the duration: ◦ replenishment of blood and muscle oxygen stores ◦ re-synthesis of ATP and PC • slow components of EPOC, the processes that occur and the duration: ◦ elevated circulation ◦ elevated ventilation ◦ elevated body temperature ◦ lactate removal and conversion to glycogen • effect of exercise intensity on EPOC and implications of the recovery process for planning exercise or training sessions.